

LEARNER WORKBOOK

FULL NAME & SURNAME

ID NUMBER:

NAME OF ASSESSOR

DATE OF ASSESSMENT

VENUE

KNOWLEDGE
SKILLS

ACHIEVED

NOT ACHIEVED

Signature of learner

Signature of Assessor

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This material has been compiled based on the **UNIT STANDARD 9013 DESCRIBE, APPLY, ANALYSE AND CALCULATE SHAPE AND MOTION IN 2-AND 3-DIMENSIONAL SPACE IN DIFFERENT CONTEXT.**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by: Training Dynamics Africa

HOW THIS LEARNER WORKBOOK IS COMPILED

This learner workbook has been compiled to provide learners towards the mathematical literacy requirements of the NQF at Level 3.

People credited with this unit standard are able to:

1. Measure, estimate, and calculate physical quantities in practical situations relevant to the adult in life or the workplace.
2. Explore describe and represent, interpret and justify geometrical relationships and conjectures to solve problems in two and three dimensional geometrical situations.

You have to study the content and guidelines given in your learner reference guide to complete this workbook.

Remember that this completed workbook must be handed in as it will form part of your formative assessment and will play a role in awarding you the required 04 credits that are to be achieved upon completion of this module.

Write neatly and legibly and ensure that you do not simply copy the information contained in the learner reference guide into your workbook. Give your own opinion and ideas where applicable and write the answers in your own words to indicate your level of understanding and learning.

Good luck with the completion of this workbook.

LEARNING UNIT 1

Metric System

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Explain what is meant with accuracy.
2. Explain the term rounding.
3. Do basic calculations using: addition, subtraction, multiplication, division, powers and roots.
4. Apply the measurement system and instruments used in South Africa as set out in this guide.
5. Apply area and volume calculation within an operational environment.
6. Compile a rough sketch of an area or scene of incident.
7. Interpret a plan drawn according to scale.

LEARNER ACTIVITY 1

Convert the following using the metric system:

a) 35 m = _____ mm

b) 0,7 m = _____ cm

c) 56 mm = _____ m

d) 80 km = _____ m

e) 32 cm = _____ mm

f) 790 cm = _____ m

g) 38 543 g = _____ kg

h) 789 mg = _____ g

i) 2,5 l = _____ ml

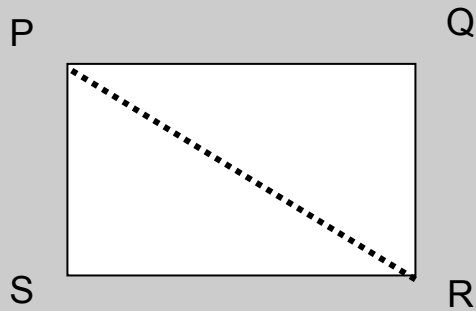
j) 90 ml = _____ l

Requirements achieved	YES		NO	
Comments in support of decision				

LEARNER ACTIVITY 2

Use the theorem of Pythagoras to calculate the following:

- ♦ A rectangular field is to be divided along its diagonal PR into two plots. If the dimensions of the field are $150\text{m} \times 75\text{m}$ calculate the length of the fencing needed to divide the field (length of PR)
- ♦ Round off to an acceptable level of accuracy.

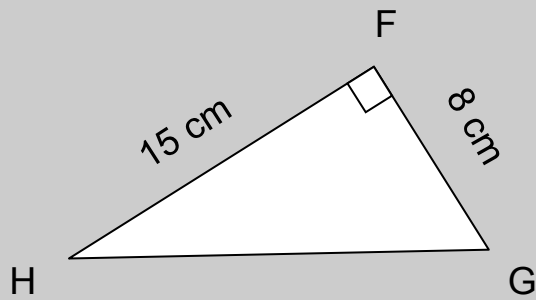


Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 3

Calculate the length of side GH in $\triangle FGH$.

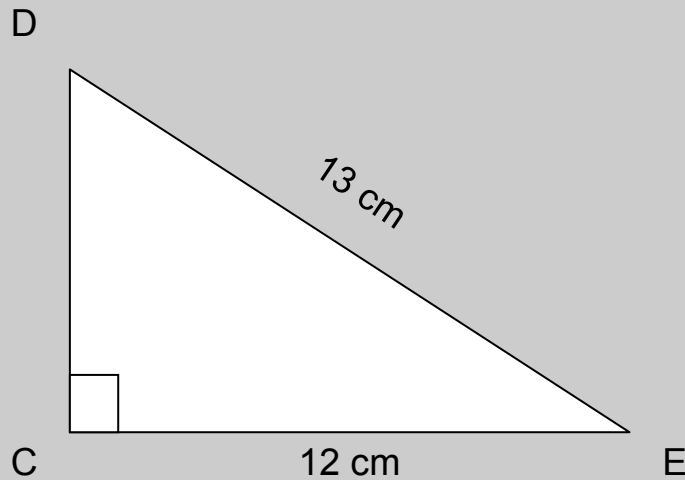


Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 4

Find the length of side CD in $\triangle CDE$.



Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 5

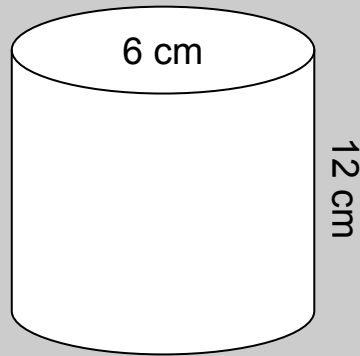
Use the formulae of perimeter, area, volume and surface area to calculate the following:

- ◆ Find the volume of a box with a rectangular base measuring 20 cm by 5 cm and a height of 25 cm. (Draw a rough sketch of the box).
- ii) How many of these boxes can be packed into a crate measuring 50 cm by 35 cm by 40 cm?
- iii) Find the surface area of the box (draw a net of it first).

LEARNER ACTIVITY 6

A cylindrical can has a height of 12 cm and a diameter of 6 cm. Find:

- i) its volume
- ii) its surface area (draw a net of it first)

[illegible]

[illegible]

Requirements achieved	YES		NO	
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Comments in support of decision

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Explain what is meant with accuracy.			
2	Explain the term rounding.			
3	Do basic calculations using: addition, subtraction, multiplication, division, powers and roots.			
4	Apply the measurement system and instruments used in South Africa as set out in this guide.			
5	Apply are and volume calculation within an operational environment.			
6	Compile a rough sketch of an area or scene of incident.			
7	Interpret a plan drawn according to scale.			

Evidence Collector Signature

Date

Security Officer Signature

Date